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Aryabhatta Knowledge University, Patna

B.Tech 3rd Semester Exam., 2018

Time: 03 Hours

Code: 011305 ***Subject: ENGINEERING GEOLOGY*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option from the following (any seven): **[14]**

(a) S-wave does not pass through
- (i) sedimentary rocks
- (ii) ore bodies
- (iii) liquids
- (iv) solidified igneous masses

(b) The record of propagation of seismic waves as recorded at the specific purpose station is called
- (i) seismograph
- (ii) seismogram
- (iii) S-P travel-time curve
- (iv) None of the above

(c) An aquifer is a rock formation which is
- (i) porous and not necessarily permeable
- (ii) porous and essentially permeable
- (iii) highly porous and impermeable
- (iv) None of the above

(d) Kankar is a variety of
- (i) sandstone
- (ii) limestone
- (iii) shale
- (iv) coal

(e) Rocks are _____ so far as their elastic constants are considered.
- (i) isotropic
- (ii) anisotropic
- (iii) partly isotropic
- (iv) None of the above

(f) The capacity of a stone to withstand scratching action is defined as
- (i) strength
- (ii) hardness
- (iii) durability
- (iv) None of the above

(g) In a natural soil profile, the rock zone will be
- (i) at the surface
- (ii) somewhere in middle
- (iii) at the base
- (iv) absent

(h) Loss angles test is used for
- (i) toughness test
- (ii) abrasion test
- (iii) attrition test
- (iv) density test

(i) A good quality stone must absorb water less than
- (i) 2-5%
- (ii) 5%
- (iii) 10%
- (iv) 20%

(j) Stone generally used for railway ballast is
- (i) sandstone
- (ii) limestone
- (iii) marble
- (iv) basalt or trap basalt

Q.2 Solve both questions : **[14]**

(a) Define engineering geology. Add a note on its importance in civil engineering.

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(b) What is meant by weathering of rocks? Describe in detail the type of rock weathering.

Q.3 Solve this question : **[14]**

What are sedimentary rocks? How are they formed? Describe in detail the classification of sedimentary rocks. Add a note on the engineering and economic significance of sedimentary rocks.

Q.4 Solve both questions : **[14]**

(a) Describe the geology around Bhakra-Nangal project. Discuss how geological problems were controlled during the dam construction.

(b) Describe the few examples to explain the purpose of underground installations.

Q.5 Solve both questions : **[14]**

(a) Define earthquake. Give a detailed account of origin of tectonic earthquake and their distribution on globe.

(b) Write an essay on internal and external cause of slides as they are commonly understood.

Q.6 Solve both questions : **[14]**

(a) Give a note about remote sensing and briefly explain the different types of remote sensing.

(b) Give an account of different methods used for exploration of groundwater reserves.

Q.7 Solve this question : **[14]**

With the aid of diagrams, describe the procedures in determining rock strength using the following tests:
(a) Point-load index test
(b) Triaxial compressive test
(c) Tensile strength by Brazilian test
(d) Abrasive resistance by Dory's abrasion testing machine

Q.8 Solve both questions : **[14]**

(a) Discuss the objects and general methods of geological investigation for major engineering projects.

(b) Write an essay on 'soil erosion-causes and control'.

Q.9 Write short notes on any four of the following: **[14]**

(a) Crystallography
(b) Outline of stratigraphy of India
(c) Impact test
(d) Formation of igneous rock with suitable example
(e) Difference between groundwater and artesian water
(f) Assessment of seismic risks

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